

6	(a) (i) $E = V / d$ $= 350 / (2.5 \times 10^{-2})$ $= 1.4 \times 10^4 \text{ N C}^{-1}$	C1	A1	[2]
	(ii) force = Eq $= 1.4 \times 10^4 \times 1.6 \times 10^{-19}$ $= 2.24 \times 10^{-15}$	C1	M1	A0
	(b) (i) $F = ma$ $a = (2.24 \times 10^{-15}) / (9.1 \times 10^{-31})$ $= 2.46 \times 10^{15} \text{ m s}^{-2}$...(<i>allow</i> 2.5×10^5)	C1	A1	[2]
	(ii) $s = \frac{1}{2}at^2$ $2.5 \times 10^{-2} = \frac{1}{2} \times 2.46 \times 10^{15} \times t^2$ $t = 4.5 \times 10^{-9} \text{ s}$	C1	A1	[2]
	(c) <i>either</i> gravitational force is normal to electric force <i>or</i> electric force horizontal, gravitational force vertical <i>special case:</i> force/acceleration due to electric field >> force/acceleration due to gravitational field, allow 1 mark	B2		[2]

First variant Mark Scheme

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